Quantification: 877_wt_z087c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	13
interior lacuna count	9
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 9)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	3532.86	3114	2023.13	px^2
major_axis_length_px	lacuna shape	98.6806	93.6668	25.9271	px
minor_axis_length_px	lacuna shape	46.6071	46.7845	15.9389	px
aspect_ratio	lacuna shape	2.2368	2.0448	0.5814	unitless
eccentricity	lacuna shape	0.8771	0.8723	0.0517	unitless
solidity	lacuna shape	0.7984	0.8167	0.0754	unitless
roots_count	headline	10.6667	13	6.9101	count
ring_length_r30_px	headline	374.889	374	221.574	px
ring_length_r60_px	reported	879.222	1029	490.792	px
owned_length_px	ownership-dependent	1837.41	1888.16	1252.07	px
edge_count	ownership-dependent	87.7778	97	61.8441	count
mean_edge_length_px	ownership-dependent	20.5552	20.5109	4.2631	px
perimeter_px	normalised	295.653	301.404	104.145	px
ring_area_r30_px2	normalised	9718.33	9757	3317.86	px^2
ring_area_r60_px2	normalised	23702.2	24715	7555.84	px^2
in_frame_fraction_r30	normalised	0.9994	1	0.0019	unitless
in_frame_fraction_r60	normalised	0.9845	1	0.0465	unitless
ring_density_r30	normalised	0.0373669	0.0428194	0.0152515	px^-1
ring_density_r60	normalised	0.0363428	0.043519	0.0161201	px^-1
roots_per_100px_perimeter	normalised	3.4769	4.1357	1.9486	per 100 px
ring_attached_length_r30_px	network v2	344	349	222.787	px
ring_attached_length_r60_px	network v2	744.778	900	482.491	px
ring_length_w_r30_px	network v2	431.669	422.777	255.429	px
ring_length_w_r60_px	network v2	1010.89	1160.41	568.493	px
sholl_crossings_r10	network v2	9	9	5.7446	count
sholl_crossings_r20	network v2	11.5556	10	7.0907	count
sholl_crossings_r30	network v2	12.4444	14	7.002	count
loop_count	loops, report only	7.7778	9	6.2205	count
loop_edge_share	loops, report only	0.2415	0.2561	0.1539	share
ring_width_mean_r30_px	width	6.7824	6.8628	0.8571	px

Per field
----

measure	value	units
analysed_area_px2	1.009e+06	px^2
total_skeleton_length_px	31590	px
canalicular_length_density_per_px	0.0313084	px^-1
junction_count	657	count
junction_density_per_px2	0.00065114	px^-2
lacuna_count	13	count

lacunae_per_px2	1.288e-05	px^-2
skeleton_component_count	375	count
mean_component_length_px	84.24	px
median_component_length_px	23	px
field_length_density_w_per_px	0.0358004	px^-1
field_density_without_flagged_per_px	0.0313084	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	700	px
unattached_component_count	35	count
canalicular_length_density_attached_per_px	0.0313084	px^-1
field_length_density_with_unattached_per_px	0.0320021	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	94	count
loop_edge_share	0.2035	share
edge_count	1651	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.3681	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	31590	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1757	77.0533	35.9762	2.1418	0.8843	0.7142	1.417	13.6949	531.608	True
2	4317	116.364	50.6241	2.2986	0.9004	0.8545	0.8274	37.7415	814.323	True
3	1047	73.2742	21.411	3.4223	0.9564	0.675	0.4727	102.495	237.658	False
4	5980	133.077	65.0796	2.0448	0.8723	0.7429	0.7964	208.483	500.988	False
5	4142	119.005	48.2749	2.4652	0.914	0.8826	0.2092	399.882	20.3518	True
6	4805	118.959	55.0153	2.1623	0.8866	0.8357	0.4696	413.392	307.567	False
7	3114	93.6668	46.7845	2.0021	0.8663	0.8167	0.8128	455.325	614.1	False
8	5891	121.42	63.3439	1.9168	0.8531	0.8972	1.4767	461.976	812.6	False
9	1692	71.502	34.0722	2.0985	0.8792	0.9082	0.0325	636.572	12.6684	True
10	1983	66.6089	40.8008	1.6325	0.7904	0.8555	1.3986	723.423	816.988	False
11	1910	83.7583	33.8144	2.477	0.9149	0.7658	1.3629	757.078	851.197	False
12								860.708	550.499	False
13								970.308	685.063	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	10	377	769	1999.58	101	19.7978	7.4942
2	9	285	628	1076.92	42	25.6409	6.8411
3	13	374	1029	1888.16	97	19.4656	8.1788
4	22	753	1615	3299.72	162	20.3687	7.6224

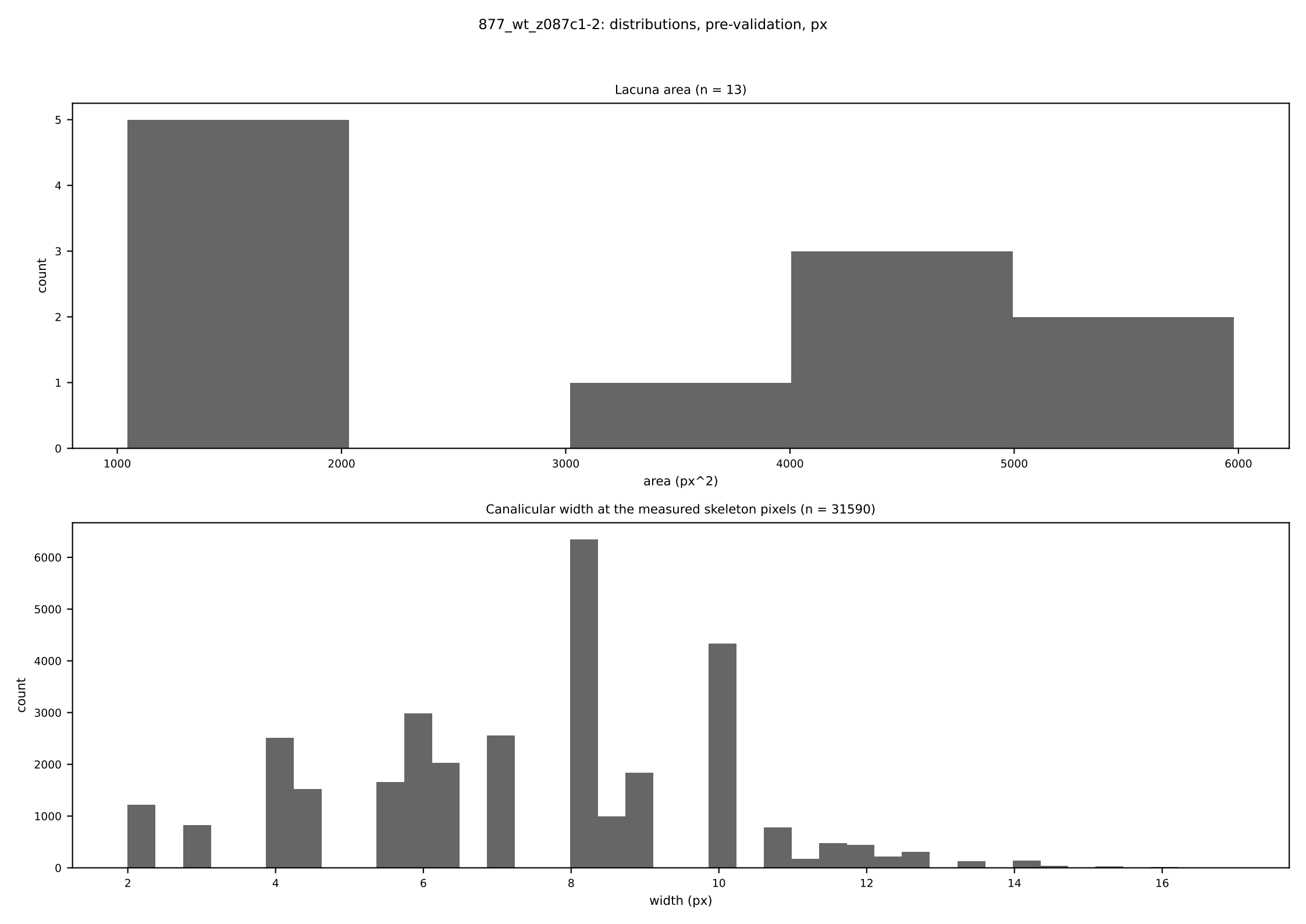
5	7	259	627	795.968	45	17.6882	6.8736
6	16	521	1200	2543.66	133	19.1252	7.2367
7	13	510	1250	3363.79	164	20.5109	6.3544
8	13	519	1124	2815.01	121	23.2645	7.0088
9	6	158	454	509.546	22	23.1612	7.8262
10	6	243	603	953.507	39	24.4489	5.7966
11	11	271	694	1323.7	58	22.8224	6.8628
12	1	109	315	317.392	13	24.4148	6.5147
13	1	74	83	31.7279	3	10.576	5.4666

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	274.066	6033	14351	0.6294	0.5895	0.0624896	0.0535851	3.6488
2	340.475	9033	20414	0.7674	0.7052	0.031551	0.0307632	2.6434
3	206.551	8300	22062	1	1	0.0450602	0.0466413	6.2938
4	502.186	15464	35653	1	1	0.0486937	0.0452977	4.3808
5	336.628	7317	17241	0.6107	0.5856	0.035397	0.0363668	2.0794
6	357.404	12357	30057	1	1	0.0421623	0.0399241	4.4767
7	314.333	10425	25683	1	1	0.0489209	0.0486703	4.1357
8	364.718	12512	29886	1	1	0.0414802	0.0376096	3.5644
9	190.953	4836	12429	0.5823	0.5596	0.0326716	0.0365275	3.1421
10	215.522	5675	13856	1	1	0.0428194	0.043519	2.7839
11	232.936	5964	14531	1	1	0.0454393	0.04776	4.7223
12	301.404	9757	24715	1	1	0.0111715	0.0127453	0.3318
13	165.823	7011	16877	0.9943	0.8606	0.0105548	0.00491794	0.6031

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	377	768	436.333	888.394	10	7	10
2	256	482	335.291	728.524	8	10	11
3	349	965	422.777	1160.41	9	10	14
4	727	1483	867.08	1870.84	16	21	22
5	207	371	294.279	710.784	8	6	7
6	491	1070	614.24	1394.82	14	19	20
7	468	1091	589.701	1443.47	13	16	14
8	474	900	585.387	1280.66	15	18	19
9	156	359	181.61	507.718	5	4	8
10	243	470	278.037	695.169	6	6	6
11	251	588	316.078	804.382	5	7	9
12	64	107	129.196	358.333	2	6	4
13	29	29	82.5269	89.9411	1	1	4


877_wt_z087c1-2: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 31590)
count
width (px)